

# AYUSHI CHADHA

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Website · GitHub · LinkedIn · Substack · X

## RESEARCH

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### When to Re-Plan: Subgoal Persistence in Hierarchical Latent Reasoning

ACCEPTED · Sole Author · [2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents @ ICML 2026](#) · Seoul, South Korea

Jan 2025 - May 2026 · [Paper](#) · [Code](#) / [SCOUT](#)

- Studied subgoal persistence in hierarchical latent reasoning, asking how long a model should preserve medium-horizon intent while multi-step computation unfolds in latent state.
- Extended HRM with a feudal-style manager-worker interface in which a high-level module emits an L2-normalised directional subgoal that persists for P worker steps; integrated the signal through learned hidden-state injection, commitment gating, and cosine-alignment supervision over each persistence window.
- Identified persistence period P as the governing control variable. At alignment weight  $\lambda=0.05$ , moderate persistence P in [3,6] outperformed both single-step re-planning and longer commitments; P=1 fell below the no-subgoal baseline.
- Designed full, no-subgoal, and random-direction ablations to isolate interference from learned directional structure rather than added capacity or auxiliary supervision; additional sweeps characterised alignment strength, normalisation, and commitment gating.

### A Score Is Not a Policy: Measuring Whether a Supervisory Module's Decisions Are Worth Making

Sole Author · Preprint · Under Review

May 2026 - Present · [Paper](#)

- Extended fixed persistence to learned state-conditioned meta-control, where a supervisory controller observes the reasoner's internal trajectory and decides whether to preserve or replace an active latent commitment under matched compute.
- Evaluated learned revision policies against strong fixed schedules across precommitted seeds and calibrated intervention budgets. Despite apparently state-dependent scores, adaptive controllers did not outperform the strongest forced timing policy on their corresponding frozen checkpoints.
- Diagnosed a separation between score dependence, policy adaptation, and decision value: position alone explained ~96.8% of controller-score variance in the seed-0 diagnostic, while a time-only reconstruction reproduced the full-score ordering with Spearman  $\rho \approx 0.978$ .
- Used matched PERSIST / REPLAN counterfactuals, controlled timing interventions, and exhaustive enumeration of the K=2 revision space. The best fixed schedule captured ~71% of the gap from uniform-random timing to a per-episode oracle, with remaining adaptive headroom concentrated in ~7.6% of episodes.

## RESEARCH SERVICE & INVITED PRESENTATIONS

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### Invited Reviewer · [NeurIPS 2026 Workshop on Responsible Use of Meta-Agents](#)

Nominated by the Area Chair to review research on supervisory systems that monitor, coordinate, evaluate, or manage other agents.

### Reviewer · [2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents @ ICML 2026](#)

Reviewed research on compositional learning, reasoning, interpretability, and agentic systems.

### Invited Oral Presenter · 2nd Global Summit on Innovating the Future of Artificial Intelligence (InnoAI-2027)

Kuala Lumpur, Malaysia · April 2027

PROFESSIONAL EXPERIENCE

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| <b>Propero Consulting</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Pune, India         |
| Software Developer · Machine Learning & Applied AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Aug 2022 - Dec 2024 |
| <ul style="list-style-type: none"><li>Owned the architecture and development of ShopiBot, a domain-specific AI system for Shopify developers supporting technical question answering, requirements understanding, implementation guidance, and agentic workflows.</li><li>Developed retrieval and reasoning pipelines using dense retrieval, RAPTOR, corrective/adaptive RAG, LangChain, vector-database sharding, and prompt/embedding caching; reduced end-to-end latency from 53s to 20s (~62%), improved top-k retrieval precision by 28%, and reduced hallucinations by 35%.</li><li>Developed agentic workflows using knowledge graphs, LangChain planners, multi-tool reasoning, and self-reflection; subsequently experimented with Sema4AI / Robocorp and LLM tool calling for domain-aware developer automation.</li><li>Built multimodal generation systems using CLIP-guided prompt augmentation, DreamBooth, and Stable Diffusion; also developed automated evaluation and testing systems, including an anomaly-aware checkout-flow tester that caught 92% of regressions pre-release.</li></ul> |                     |
| <b>Unify · YC-backed</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | London / Remote     |
| LLM Engineer Contributor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jul 2024 - Sep 2024 |
| <ul style="list-style-type: none"><li>Selected for Unify's contributor programme, collaborating with the core team on a community-built unified LLM query router across model providers.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |

SAINT

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| <b>Saint</b>                                                                                                                                                                                                                                                                                                                                                                                      | 2026 - Present |
| Independent Venture                                                                                                                                                                                                                                                                                                                                                                               |                |
| <ul style="list-style-type: none"><li>Developing evaluation methods for agentic and reasoning systems to measure whether additional computation, retries, or strategy changes improve downstream outcomes.</li><li>Grounded in research on subgoal persistence and supervisory decision value, with initial applications to coding-agent repair loops and research-agent investigation.</li></ul> |                |

EDUCATION

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|-------------------------------------------------------------------------------------|--------------------|
| <b>Northeastern University</b>                                                      | Boston, MA         |
| M.S. Artificial Intelligence · Machine Learning concentration                       | Admitted Fall 2026 |
| Khoury College of Computer Sciences · Spring 2027 start option                      |                    |
| International Impact Award · Inaugural Cohort Scholarship · 30% tuition scholarship |                    |
| <b>G.B. Pant University of Agriculture &amp; Technology</b>                         | Pantnagar, India   |
| B.Tech, Electrical Engineering · 73.2% · First Division                             | 2017 - 2021        |

TECHNICAL SKILLS

Reasoning & ML: Hierarchical and recurrent latent reasoning, adaptive computation, supervisory / meta-level control, test-time computation, hierarchical and feudal RL, PyTorch, Hugging Face Transformers

LLM & Agentic Systems: Dense retrieval, RAPTOR, corrective/adaptive RAG, LangChain, LlamaIndex, tool calling, LangSmith, CLIP, diffusion models

Engineering & Infrastructure: Python, C, C++, CUDA, Docker, AWS (S3, Lambda, EKS), Pinecone, Chroma, mixed precision, gradient checkpointing, single-node multi-GPU training